GCC Soil Water Security Simulator

Executive Summary — where did the water go?

> *Retention curves tell us how water is **stored**. Water balances tell us whether water is **saved**. Until the fluxes are measured, water-security claims are hypotheses, not outcomes.*

The problem

Soil amendments are sold on one claim: they make soil hold more water — and from that, vendors leap to “reduced irrigation” and “water security.” That leap is not scientifically valid. **Holding more water is not the same as using less water.** A soil can store more and still lose it to evaporation, drainage, or extra plant uptake, leaving the irrigation bill unchanged. In the Gulf, where evaporative demand is extreme, this gap is the rule, not the exception.

What this platform does differently

Instead of stopping at “how much can the soil hold?”, GCC-SWSS runs a complete daily water balance over a full year and tracks every millimetre of rain and irrigation to its fate: plant uptake (useful), evaporation, drainage, runoff, or storage. It does this for an unamended baseline **and** each amendment, comparing them at **equal plant health** — a fair comparison, not “healthy plant vs stressed plant.”

A typical finding (real model output)

For a Dubai-climate landscape (sandy loam, warm-season turf), adding biochar:

What the vendor measures	What actually happens to the water budget
Available water storage **+13%**	Annual irrigation requirement –**0.2%**

Why almost no saving despite a real +13% storage gain? Because roughly **94% of all applied water** leaves as plant transpiration regardless of the amendment, and the soil was already meeting the plant’s needs. The extra storage capacity goes largely unused. The platform reports this honestly instead of converting the 13% into a marketing claim.

Strength of evidence, made explicit

Amendment	Evidence for water effects
Biochar, Compost	High
Biosolids, Engineered biochar, Polymer	Medium
Lignite nanocarbon (LNC), Dust-suppression biopolymer	Low

What it can — and cannot — say

It can: screen amendments and climates quickly; show where water goes; compare options on irrigation, drainage, water security, payback and ROI; and put honest uncertainty on every number. **It cannot (yet):** replace field measurement. The physics is verified and internally consistent — the water balance conserves mass exactly and the climate maths match the FAO international reference — but it has not yet been calibrated against a specific Gulf site. For a site-specific guarantee, the model is calibrated against field data (the planned next phase). Stating this plainly is what makes the tool credible rather than another marketing instrument.

Bottom line

> *GCC-SWSS replaces an unprovable marketing claim (“holds more water”) with a defensible engineering question (“how much irrigation will I actually avoid, and what happens to the rest?”). In the Gulf, the honest answer is often “less than you were told” — and knowing that before you buy is exactly the point.*